THRIVING SCHOLARS

ESAT 2026 Extra Practice Pack

20 Questions
No Calculator
Suggested time: 30 minutes

Design note: These are original questions, not recalled or reconstructed live-test items. They are calibrated to broad public candidate trends from the first two days of the October 2026 sitting and kept within the official ESAT content specification.

Student instructions: Select one answer, A–E, for each question. Work efficiently: the real ESAT allows about 89 seconds per question on average. If a route is becoming too long, use elimination and move on.

Mathematics 1 — Questions 1–5

Varied, fast application: probability, scaling, bounds, proportion and bearings.

1 A bag contains 4 red, 3 blue and 2 green counters. Two counters are drawn at random without replacement. Given that at least one of the counters drawn is red, what is the probability that the two counters are the same colour?

A 1/6

B 3/13

C 1/4

D 4/13

E 5/18

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2 Two solid models are mathematically similar. The ratio of their surface areas, smaller : larger, is 9 : 25. Their combined volume is 1216 cm^3. What is the volume of the larger model?

A 864 cm^3

B 936 cm^3

C 1000 cm^3

D 1064 cm^3

E 1125 cm^3

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3 The length and width of a rectangle are recorded as 12 cm and 8 cm respectively, each correct to the nearest centimetre. If its true area is A cm^2, which interval must contain A?

A $86 \le A < 106$

B $86.25 \le A < 106.25$

C $87.5 \le A < 105.5$

D $88 \le A \le 106$

E $90 \le A < 104$

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4 For positive x, the quantity y is inversely proportional to x^2. When $x = 3$, $y = 8$. What is x when $y = 18$?

A 3/2

B 2

C 9/4

D 5/2

E 3

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5 From point A, point B is 6 km away on a bearing of 030°. From B, point C is 8 km away on a bearing of 120°. What is the distance AC?

A 2 km

B √28 km

C 8 km

D 10 km

E 14 km

Mathematics 2 — Questions 6–10

Weighted toward algebra, expansion and geometry, reflecting the strongest early candidate trend.

6 What is the coefficient of x^3 in

$$(1 + x + x^2)^4 \text{ ?}$$

A 12

B 16

C 20

D 24

E 28

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7 The line $y = x + k$ is tangent to the circle

$$(x - 3)^2 + (y + 2)^2 = 25.$$

There are two possible values of k. What is their sum?

A −10

B −5

C 0

D 5

E 10

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8 The line $y = mx + 1$ is tangent to the parabola $y = x^2 + 2x + 5$. There are two possible values of m. What is their product?

A -12 **B** -8

C -4 **D** 4

E 12

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9 A geometric sequence has positive terms. Its first term is 12 and the sum of its first three terms is 21. What is the sum to infinity of the sequence?

A 18 **B** 21

C 24 **D** 27

E 30

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10 The polynomial $P(x) = x^3 + ax^2 + bx + 6$ is divisible by both $x - 1$ and $x + 2$. What is $P(4)$?

A 6 **B** 12

C 18 **D** 24

E 30

Physics — Questions 11–14

Extra electricity, plus mechanics and matter; designed around quick principle selection followed by calculation.

11 A 6 Ω resistor and a 3 Ω resistor are connected in parallel across a 12 V battery. How much charge passes through the battery in 15 s?

A 30 C **B** 45 C

C 60 C **D** 90 C

E 180 C

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12 An ideal transformer has a primary voltage of 240 V. Its secondary supplies 12 V at 4.0 A. How much charge passes through the primary coil in 2.0 minutes?

A 12 C **B** 24 C

C 48 C **D** 96 C

E 480 C

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13 A car of mass 1200 kg slows uniformly from 20 m s^{-1} to 10 m s^{-1} while travelling 45 m. What is the magnitude of the resultant braking force?

A 3000 N

B 3600 N

C 4000 N

D 4500 N

E 6000 N

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14 A fixed amount of gas occupies 600 cm^3 at a pressure of 100 kPa. It is compressed at constant temperature to 250 cm^3. The gas then pushes on a piston of area 5.0 cm^2. What force does the gas exert on the piston?

A 12 N

B 48 N

C 120 N

D 240 N

E 480 N

Biology — Questions 15–17

Short stems with close conceptual distinctions rather than long passages.

15 Three identical potato cylinders are placed for the same time in solutions X, Y and Z. The cylinder in X gains mass, the cylinder in Y shows no change in mass, and the cylinder in Z loses mass. Which ordering of the solutions' water potentials is consistent with these results?

A $\Psi_X > \Psi_Y > \Psi_Z$

B $\Psi_Z > \Psi_Y > \Psi_X$

C $\Psi_X = \Psi_Z > \Psi_Y$

D $\Psi_Y > \Psi_X > \Psi_Z$

E $\Psi_X > \Psi_Z > \Psi_Y$

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16 A genetic condition is caused by a recessive autosomal allele. Two unaffected parents have an affected child. Assume no new mutation. What is the probability that their next child is unaffected but carries the recessive allele?

A 1/4

B 1/3

C 1/2

D 2/3

E 3/4

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17 Identical samples of an enzyme are kept for 10 minutes at 20°C, 40°C and 70°C. All three are then placed at 40°C and given fresh substrate. The sample previously kept at 70°C has very little activity, while the samples previously kept at 20°C and 40°C have similar high activity. Which explanation is best?

A The enzyme at 20°C was permanently denatured, but later repaired.

B High temperature permanently altered many active sites; low temperature mainly reduced collision rate temporarily.

C The enzyme at 70°C became more specific for a different substrate.

D Low temperature destroyed peptide bonds, while high temperature reduced kinetic energy.

E The 70°C sample had converted all of its enzyme into product.

Chemistry — Questions 18–20

Knowledge/application weighted: tests, equilibrium and electrolysis rather than long calculations.

18 An aqueous solution of an unknown salt gives a blue precipitate when aqueous sodium hydroxide is added. A separate sample gives a white precipitate when acidified silver nitrate is added. Which salt is the best match?

A $CuCl_2$

B $CuBr_2$

C $FeCl_2$

D $FeCl_3$

E $MgCl_2$

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19 Ammonia is formed in the reversible exothermic reaction

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g).$$

Which change increases both the equilibrium amount of ammonia and the initial rate of the forward reaction?

A Increase the pressure.

B Increase the temperature.

C Add a catalyst only.

D Decrease the concentration of nitrogen.

E Decrease the pressure.

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20 Aqueous copper(II) sulfate is electrolysed using inert graphite electrodes. Which observation is expected?

A The cathode loses mass and hydrogen forms at the anode.

B Copper is deposited at the cathode and oxygen is produced at the anode.

C Copper is deposited at the anode and oxygen is produced at the cathode.

D Hydrogen is produced at both electrodes.

E No products form because sulfate ions are not discharged.

Answers & Worked Solutions

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	6	B	11	D	16	C
2	C	7	A	12	B	17	B
3	B	8	A	13	C	18	A
4	B	9	C	14	C	19	A
5	D	10	C	15	A	20	B

1. Mathematics 1 — Conditional probability

Answer B: 3/13

P(at least one red) = 1 − P(no red) = 1 − (5/9)(4/8) = 1 − 5/18 = 13/18.
If the two counters are the same colour and at least one is red, they must both be red: P(RR) = (4/9)(3/8) = 1/6.
Required probability = (1/6) ÷ (13/18) = 3/13.

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2. Mathematics 1 — Similar solids

Answer C: 1000 cm³

Surface-area ratio 9:25 gives linear ratio 3:5. Therefore volume ratio = $3^3:5^3$ = 27:125. The total is 152 ratio-parts, corresponding to 1216 cm³, so each part is 8 cm³. Larger volume = 125 × 8 = 1000 cm³.

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3. Mathematics 1 — Error intervals

Answer B: 86.25 ≤ A < 106.25

True length: 11.5 ≤ L < 12.5. True width: 7.5 ≤ W < 8.5. Both are positive, so the minimum area is 11.5 × 7.5 = 86.25, while the upper bound is 12.5 × 8.5 = 106.25, not included.

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4. Mathematics 1 — Inverse proportion

Answer B: 2

$y = k/x^2$. Using x = 3 and y = 8 gives k = 8 × 9 = 72. For y = 18: $18 = 72/x^2$, so $x^2 = 4$. Since x is positive, x = 2.

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5. Mathematics 1 — Bearings

Answer D: 10 km

AB has direction 030° and BC has direction 120°. The direction BA is 210°, so angle ABC is 90°. Hence triangle ABC is right-angled at B: AC = $\sqrt{(6^2 + 8^2)} = \sqrt{100}$ = 10 km.

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6. Mathematics 2 — Expansion

Answer B: 16

To form x^3 from four factors $(1 + x + x^2)$: either choose x from three factors and 1 from one factor: $C(4,3) = 4$ ways; or choose x^2 from one factor, x from another and 1 from the remaining two: $4 \times 3 = 12$ ways. Total coefficient = 4 + 12 = 16.

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7. Mathematics 2 — Circle tangent

Answer A: −10

The circle has centre (3, −2) and radius 5. The line $y = x + k$ can be written $x - y + k = 0$. Tangency requires the perpendicular distance from the centre to the line to equal 5: $|3 - (-2) + k|/\sqrt{2} = 5$. Thus $|5 + k| = 5\sqrt{2}$, so $k = -5 \pm 5\sqrt{2}$. Their sum is −10.

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8. Mathematics 2 — Tangency and discriminant

Answer A: −12

At intersections: $x^2 + 2x + 5 = mx + 1$, so $x^2 + (2 - m)x + 4 = 0$. Tangency means discriminant = 0: $(2 - m)^2 - 16 = 0$. Hence $2 - m = \pm 4$, giving $m = -2$ or 6. Product = −12.

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9. Mathematics 2 — Geometric series

Answer C: 24

$12(1 + r + r^2) = 21$, so $1 + r + r^2 = 7/4$. Hence $r^2 + r - 3/4 = 0$, giving $r = 1/2$ or $r = -3/2$. Since the terms are positive, $r = 1/2$. Sum to infinity $= 12/(1 - 1/2) = 24$.

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10. Mathematics 2 — Factor theorem

Answer C: 18

Two roots are 1 and −2. For a monic cubic with constant term 6, the product of its roots is −6. If the third root is r, then $(1)(-2)r = -6$, so $r = 3$. Therefore $P(x) = (x - 1)(x + 2)(x - 3)$. $P(4) = 3 \times 6 \times 1 = 18$.

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11. Physics — Parallel circuit

Answer D: 90 C

Each resistor has 12 V across it. Currents are $12/6 = 2$ A and $12/3 = 4$ A. Total battery current = 6 A. $Q = It = 6 \times 15 = 90$ C.

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12. Physics — Ideal transformer

Answer B: 24 C

For an ideal transformer, input power = output power: $240\ I_p = 12 \times 4$. Thus $I_p = 0.20$ A. Two minutes = 120 s, so $Q = It = 0.20 \times 120 = 24$ C.

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13. Physics — Kinematics and force

Answer C: 4000 N

Use $v^2 - u^2 = 2as$: $10^2 - 20^2 = 2a(45)$, so $-300 = 90a$ and $a = -10/3$ m s^{-2}. Force magnitude = $m|a| = 1200 \times 10/3 = 4000$ N.

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14. Physics — Gas pressure and force

Answer C: 120 N

At constant temperature, $P_1V_1 = P_2V_2$: $P_2 = 100 \times 600/250 = 240$ kPa = 240000 Pa.
5.0 cm^2 = 5.0×10^{-4} m^2. Since $P = F/A$, $F = PA = 240000 \times 5.0 \times 10^{-4} = 120$ N.

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15. Biology — Osmosis and water potential

Answer A: ΨX > ΨY > ΨZ

Water enters the cylinder in X, so X has a higher water potential than the cells initially. No net mass change in Y indicates approximately equal water potentials. Water leaves the cylinder in Z, so Z has the lower water potential. Therefore ΨX > ΨY > ΨZ.

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16. Biology — Recessive inheritance

Answer C: 1/2

Two unaffected parents with an affected child must both be heterozygous: Aa × Aa. The offspring probabilities are AA = 1/4, Aa = 1/2 and aa = 1/4. An unaffected carrier is Aa, so the probability is 1/2.

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17. Biology — Enzymes

Answer B

Cooling generally slows enzyme-controlled reactions because particles move more slowly and successful collisions occur less often; this does not normally destroy the enzyme. A sufficiently high temperature can denature the protein, changing the active site permanently. The failure of the 70°C sample to recover at 40°C supports denaturation.

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18. Chemistry — Qualitative tests

Answer A: $CuCl_2$

Cu^{2+} gives a blue precipitate with aqueous sodium hydroxide. Chloride ions give a white precipitate with acidified silver nitrate. Therefore the salt is copper(II) chloride, $CuCl_2$.

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19. Chemistry — Equilibrium and rate

Answer A: Increase the pressure

The left side has 4 moles of gas for every 2 moles on the right. Increasing pressure shifts equilibrium toward the side with fewer gas particles, increasing ammonia yield. It also raises collision frequency, increasing the initial forward rate. Increasing temperature would increase rate but, because the forward reaction is exothermic, reduce equilibrium ammonia yield. A catalyst increases rate but does not change the equilibrium position.

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20. Chemistry — Electrolysis

Answer B

At the cathode, Cu^{2+} ions gain electrons and are reduced: $Cu^{2+} + 2e^{-} \rightarrow Cu$. At the inert anode, oxygen is produced from water/hydroxide ions. Therefore copper is deposited at the cathode and oxygen is evolved at the anode.

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